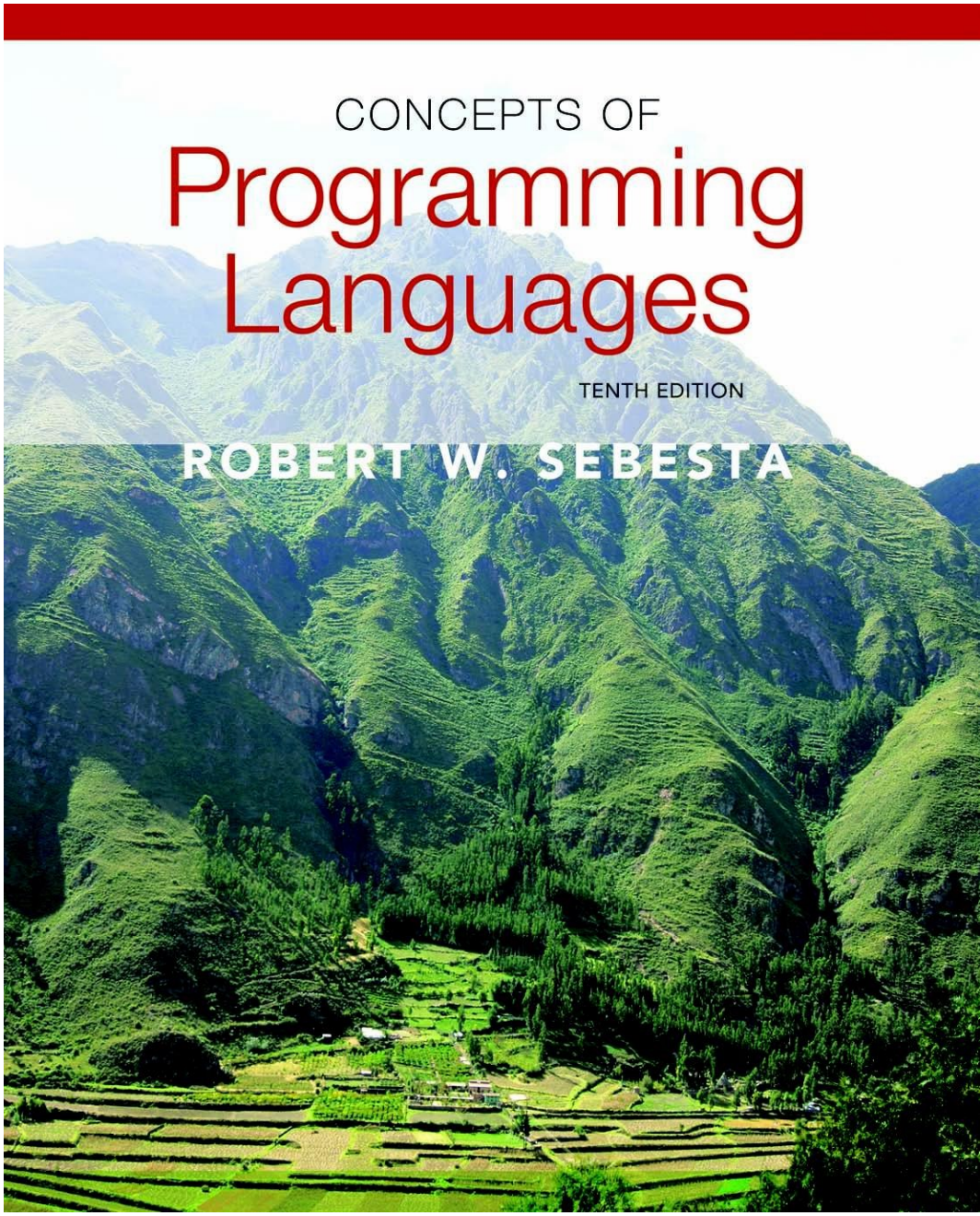
Chapter 10

Implementing Subprograms



Chapter 10 Topics

- The General Semantics of Calls and Returns
- Implementing "Simple" Subprograms
- Implementing Subprograms with Stack-Dynamic Local Variables
- Nested Subprograms
- Blocks
- Implementing Dynamic Scoping

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The General Semantics of Calls and Returns

- The subprogram call and return operations of a language are together called its *subprogram linkage*

- The implementation of subprograms must be based on the semantics of the subprogram linkage of the language being implemented

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The General Semantics of Calls and Returns

- General semantics of calls to a subprogram
  - Parameter passing methods
  - Stack-dynamic allocation of local variables
  - Save the execution status of calling program
  - Transfer of control and arrange for the return
  - If subprogram nesting is supported, access to nonlocal variables must be arranged

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The General Semantics of Calls and Returns

- General semantics of subprogram returns:
  - Out mode and in-out mode parameters must have their values returned
  - Deallocation of stack-dynamic locals
  - Restore the execution status
  - Return control to the caller

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Implementing "Simple" Subprograms

- Call Semantics:
  - Save the execution status of the caller
  - Pass the parameters
  - Pass the return address to the called
  - Transfer control to the called

"Simple" means that subprograms cannot be nested and all local variables are static.

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Implementing "Simple" Subprograms

- Return Semantics:
  - If pass-by-value-result or out mode parameters are used, move the current values of those parameters to their corresponding actual parameters
  - If it is a function, move the functional value to a place the caller can get it
  - Restore the execution status of the caller
  - Transfer control back to the caller

- Required storage:
  - Status information, parameters, return address, return value for functions, temporaries

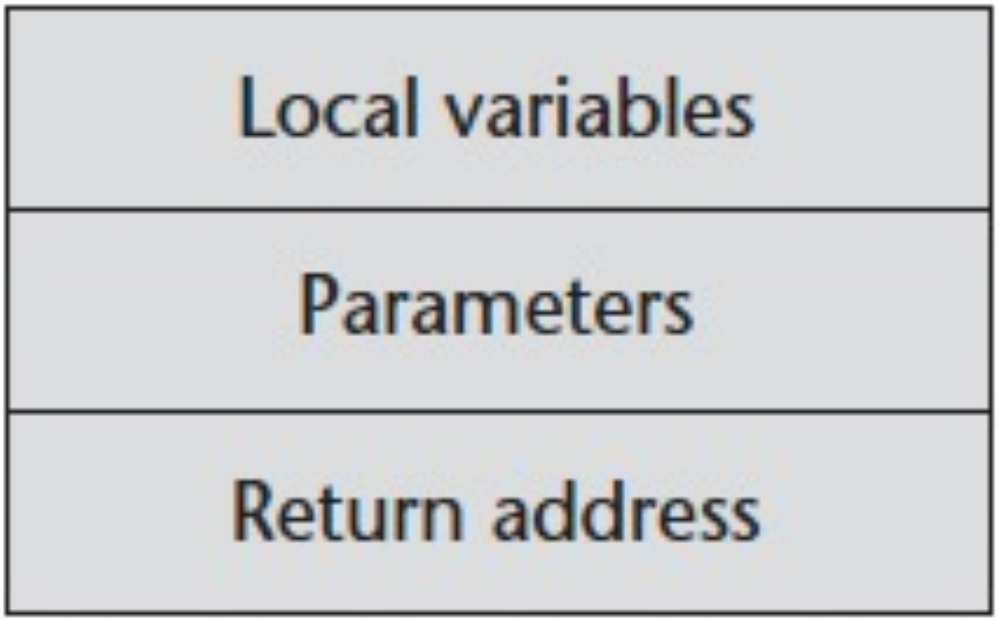
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Implementing "Simple" Subprograms

- Two separate parts:
  - the actual code and
  - the non-code part (local variables and data that can change)
- The format, or layout, of the non-code part of an executing subprogram is called an *activation record*
- An *activation record instance* is a concrete example of an activation record (the collection of data for a particular subprogram activation)

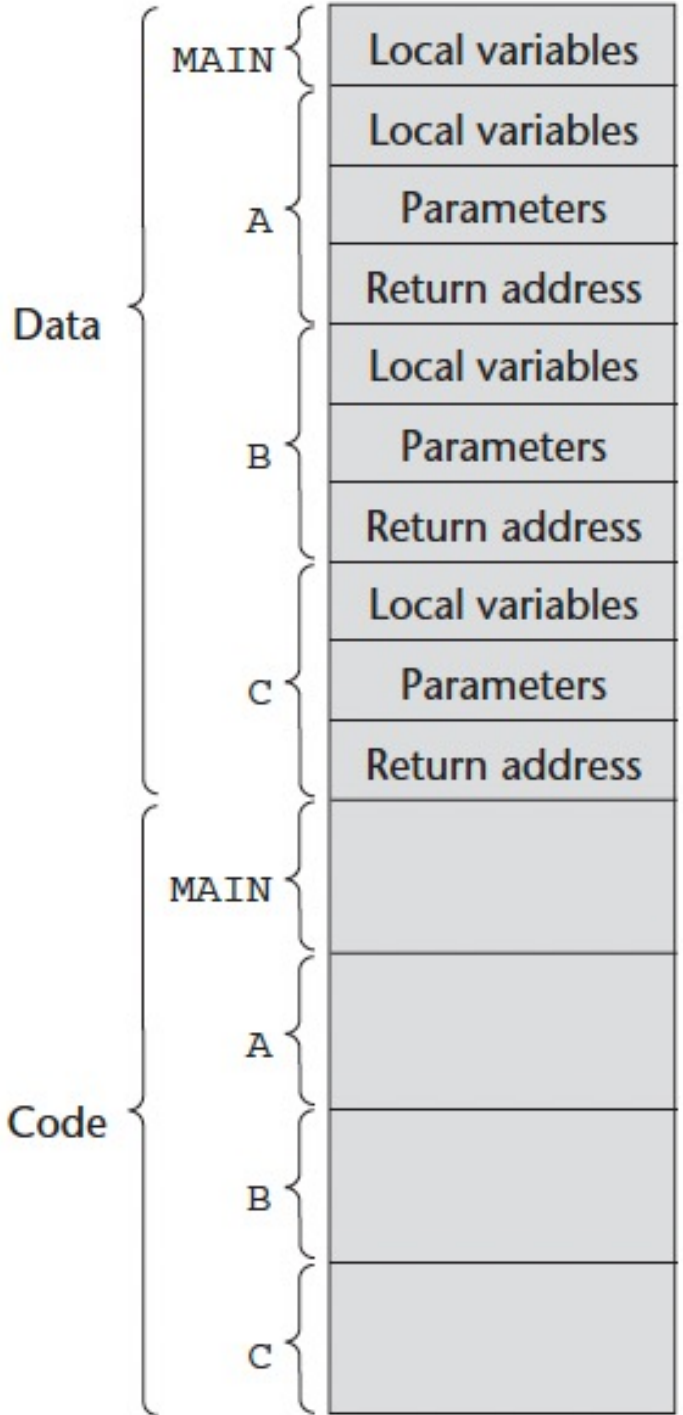
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An Activation Record for "Simple" Subprograms



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Code and Activation Records of a Program with "Simple" Subprograms



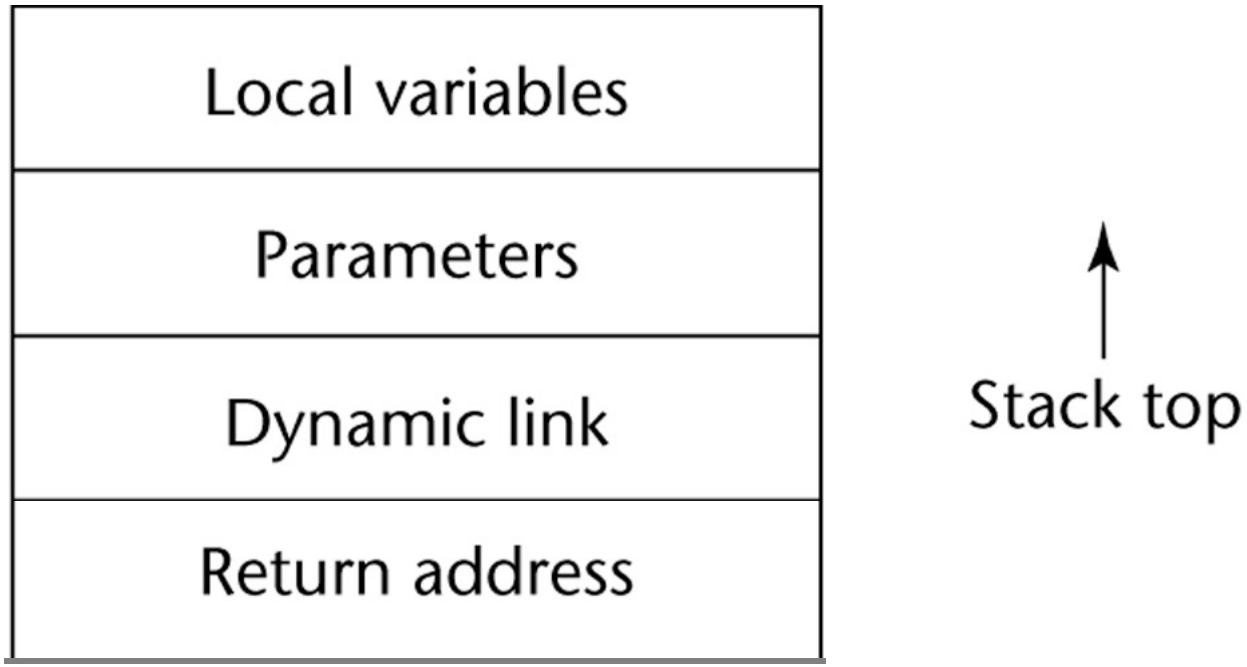
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Implementing Subprograms with Stack-Dynamic Local Variables

- More complex activation record
  - The compiler must generate code to cause implicit allocation and deallocation of local variables
  - Recursion must be supported (adds the possibility of multiple simultaneous activations of a subprogram)

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Typical Activation Record for a Language with Stack-Dynamic Local Variables



- An activation record (AR) instance is dynamically created when a subprogram is called
- AR instances reside on the run-time stack

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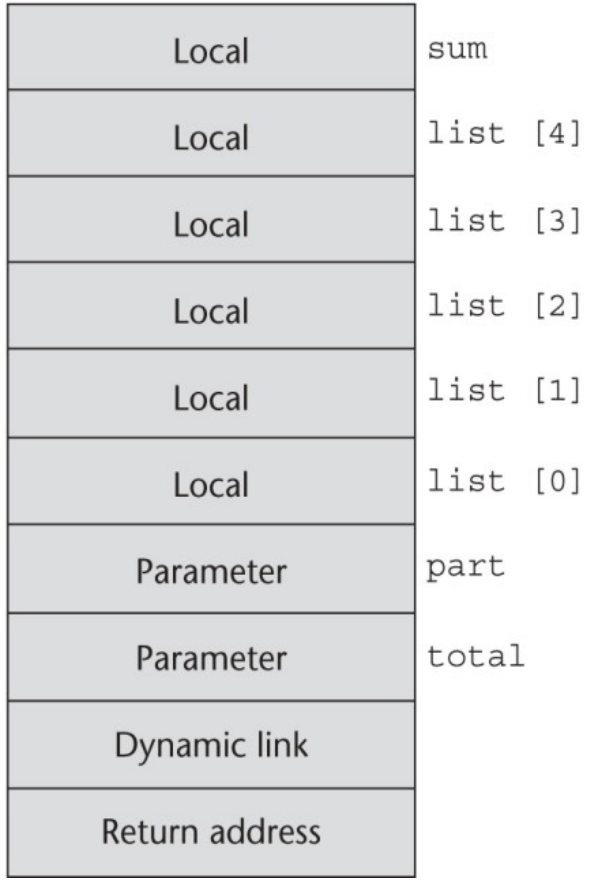
Implementing Subprograms with Stack-Dynamic Local Variables: Activation Record

- The activation record format is static, but its size may be dynamic
- The *dynamic link* is a pointer to the base of the activation record instance of the caller.
  - In static-scoped languages, dynamic link is used to provide traceback information when a run-time error occurs.
  - In dynamic-scoped languages, the dynamic link is used to access nonlocal variables.
- The *Environment Pointer* (EP) must be maintained by the run-time system. It always points at the base of the activation record instance of the currently executing program unit

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An Example: C Function

```
void sub(float total, int part){
   int list[5];
   float sum;
   …
}
```



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Revised Semantic Call/Return Actions

- Caller Actions:
  - Create an activation record instance
  - Save the execution status of the current program unit
  - Compute and pass the parameters
  - Pass the return address to the called
  - Transfer control to the called
- Prologue actions of the called:
  - Save the old EP in the stack as the dynamic link and create the new value
  - Allocate local variables

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Revised Semantic Call/Return Actions (continued)

- Epilogue actions of the called:
  - If there are pass-by-value-result or out-mode parameters, the current values of those parameters are moved to the corresponding actual parameters
  - If the subprogram is a function, its return value is moved to a place accessible to the caller
  - Restore the stack pointer by setting it to the value of the (current_EP-1) and set the EP to the old dynamic link
  - Restore the execution status of the caller
  - Transfer control back to the caller

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An Example Without Recursion

```
void fun1(float r) {
   int s, t;
   ...
   fun2(s);
   ...
}
void fun2(int x) {
   int y;
   ...
   fun3(y);
   ...
}
void fun3(int q) {
   ...
}
void main() {
   float p;
   ...
   fun1(p);
   ...
}
```



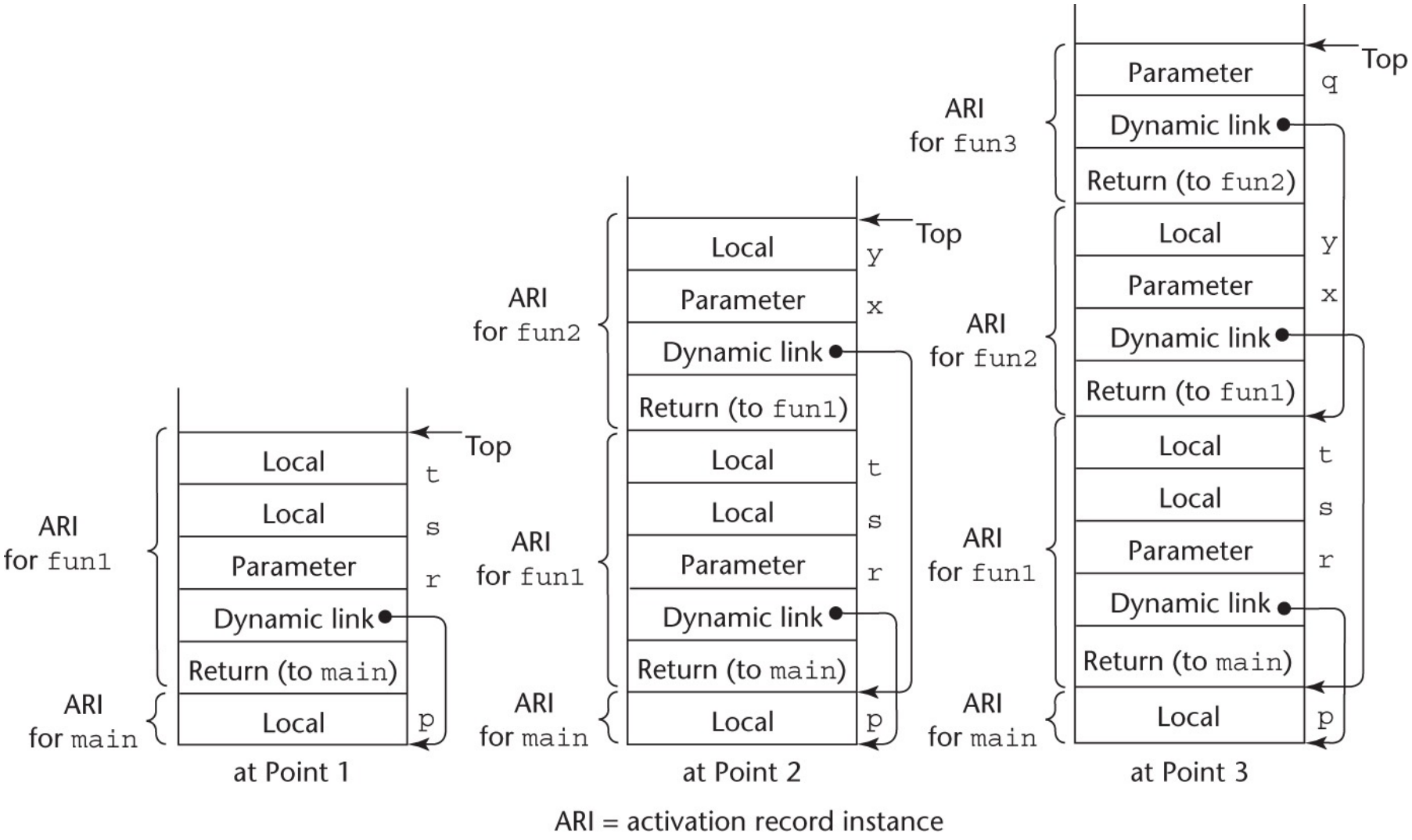




`main` **calls** `fun1`
`fun1` **calls** `fun2`
`fun2` **calls** `fun3`

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An Example Without Recursion



ARI = activation record instance

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Dynamic Chain and Local Offset

- The collection of dynamic links in the stack at a given time is called the *dynamic chain*, or *call chain*
- Local variables can be accessed by their offset from the beginning of the activation record, whose address is in the EP. This offset is called the *local_offset*
- The local_offset of a local variable can be determined by the compiler at compile time

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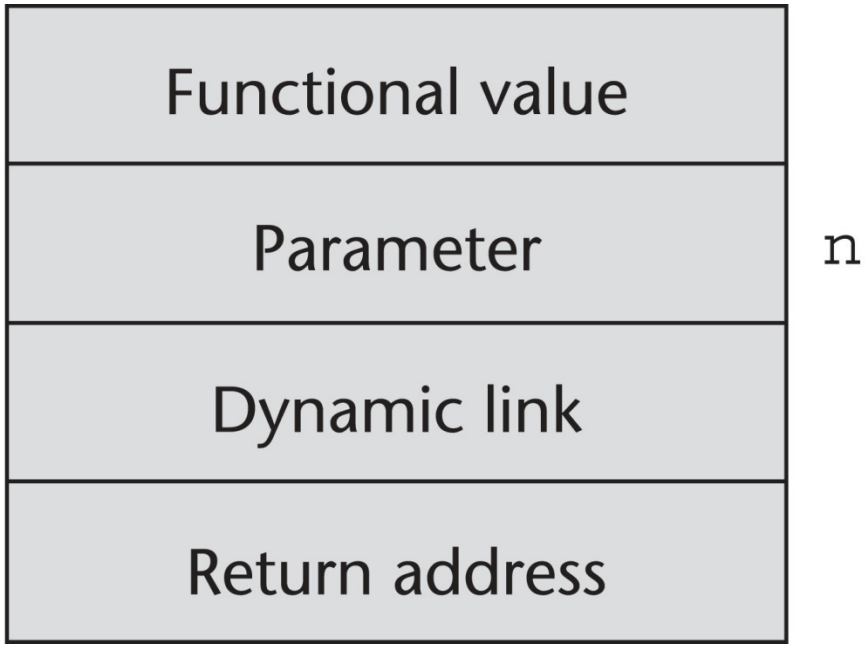
An Example With Recursion

The activation record used in the previous example supports recursion

```
int factorial (int n) {
    <------------------------------1
  if (n <= 1) return 1;
  else return (n * factorial(n - 1));
    <------------------------------2
}
void main() {
  int value;
  value = factorial(3);
    <------------------------------3
}
```

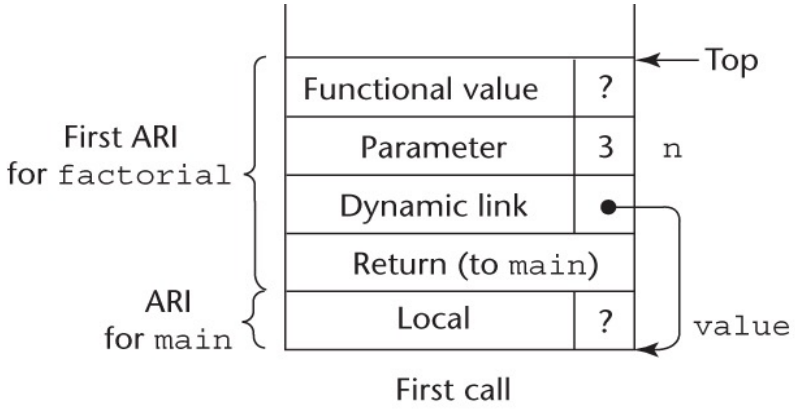
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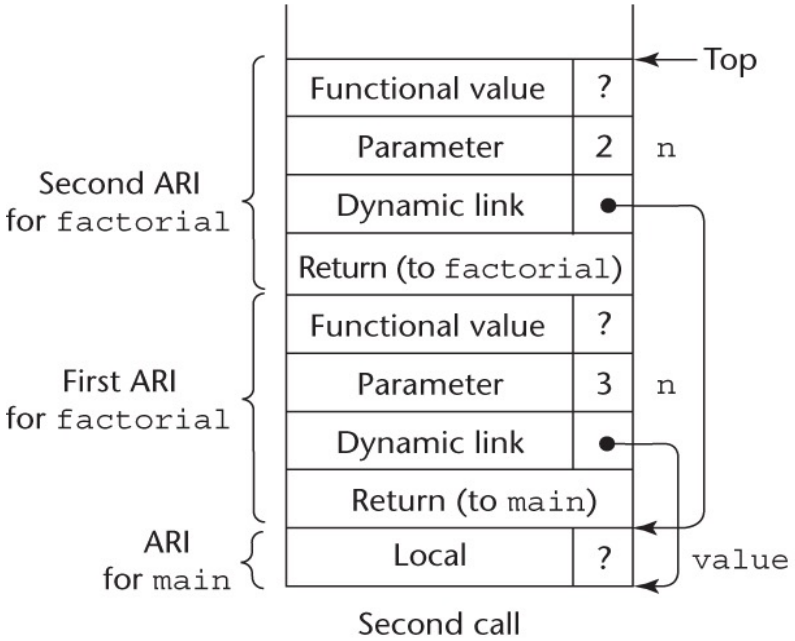
Activation Record for `factorial`

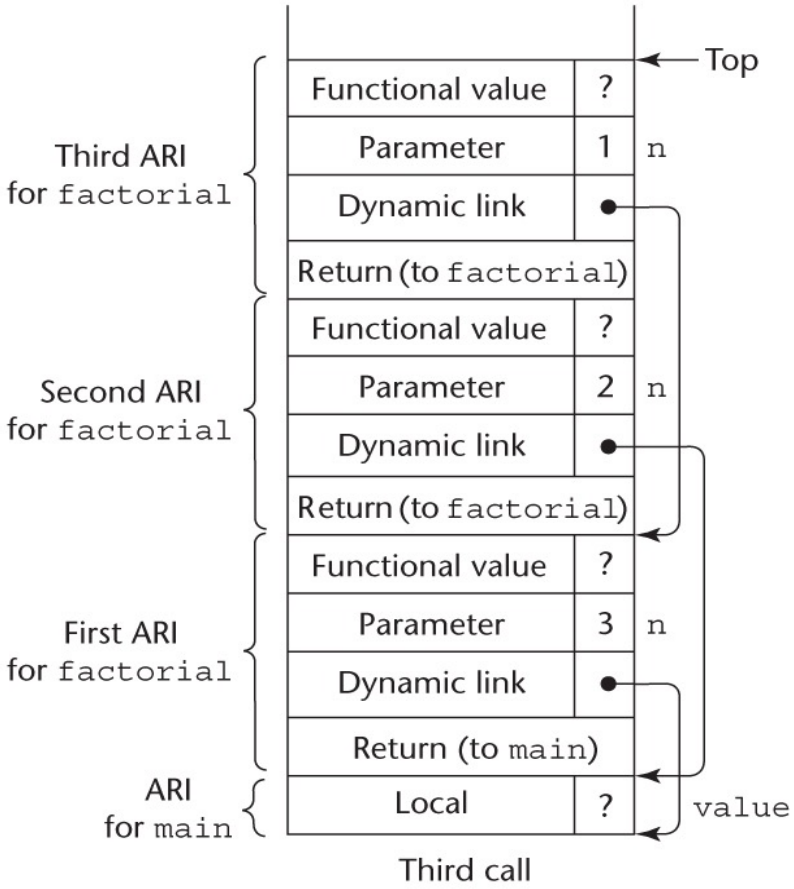


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Stacks for calls to `factorial`

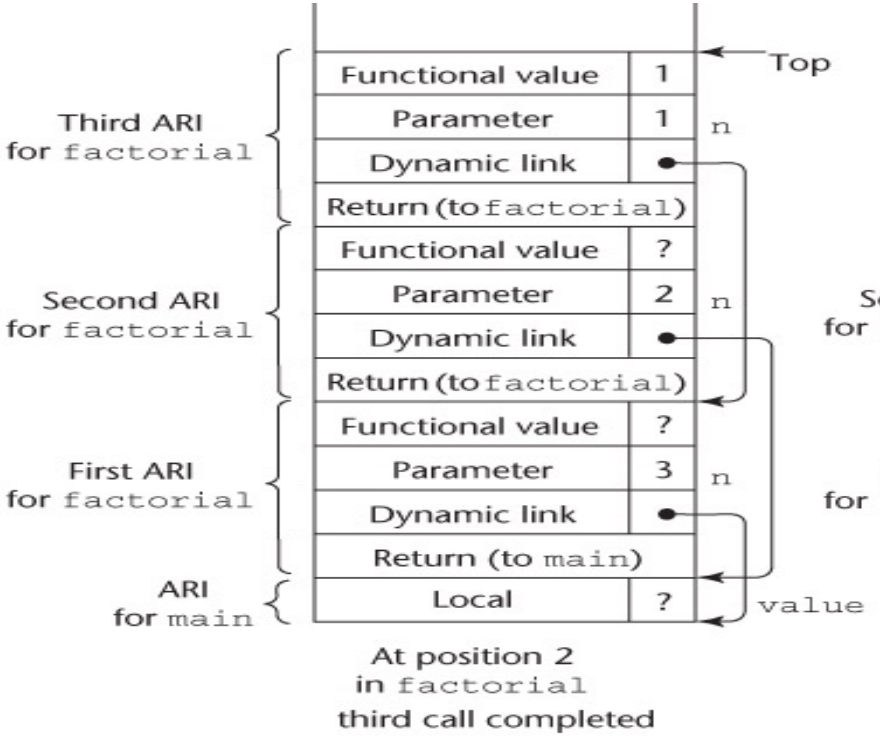




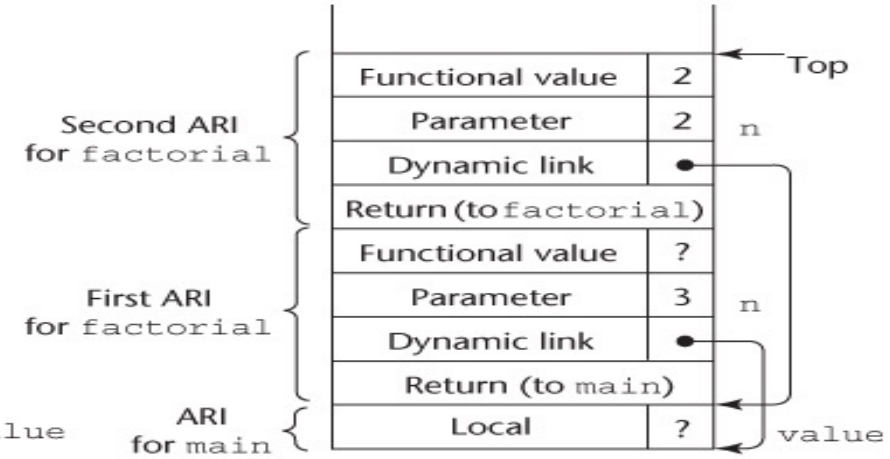


ARI = activation record instance

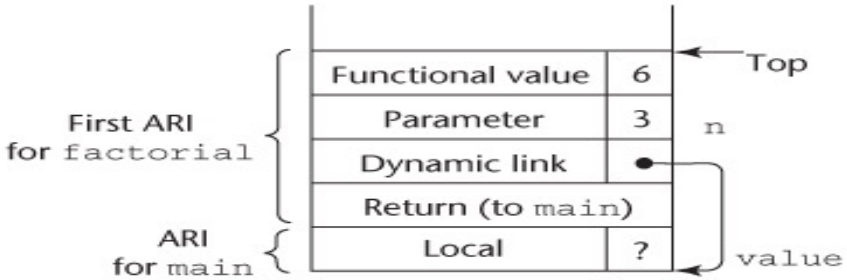
Stacks for returns from `factorial`



At position 2
in `factorial`
third call completed



At position 2
in `factorial`
second call completed



At position 2
in `factorial`
first call completed



In position 3
in `main`
final results

ARI = activation record instance

Nested Subprograms

- Some non-C-based static-scoped languages (e.g., Fortran 95+, Ada, Python, JavaScript, Ruby, and Lua) use stack-dynamic local variables and allow subprograms to be nested
- All variables that can be non-locally accessed reside in some activation record instance in the stack
- The process of locating a non-local reference:
  1. Find the correct activation record instance
  2. Determine the correct offset within that activation record instance

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Locating a Non-local Reference

- Finding the offset is easy
- Finding the correct activation record instance
  - Static semantic rules guarantee that all non-local variables that can be referenced have been allocated in some activation record instance that is on the stack when the reference is made

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Static Scoping

- A *static chain* is a chain of static links that connects certain activation record instances
- The *static link* in an activation record instance for subprogram A points to one of the activation record instances of A's static parent
- The static chain from an activation record instance connects it to all of its static ancestors
- *Static_depth* is an integer associated with a static scope whose value is the depth of nesting of that scope

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Static Scoping (continued)

- The *chain_offset* or *nesting_depth* of a nonlocal reference is the difference between the static_depth of the reference and that of the scope when it is declared

- A reference to a variable can be represented by the pair:
  (chain_offset, local_offset),
  where local_offset is the offset in the activation record of the variable being referenced

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Skeletal Python program

```
# Global scope
. . .
def f1():
    def f2():
        def f3():
            . . .
        # end of f3
        . . .
    # end of f2
    . . .
# end of f1
```

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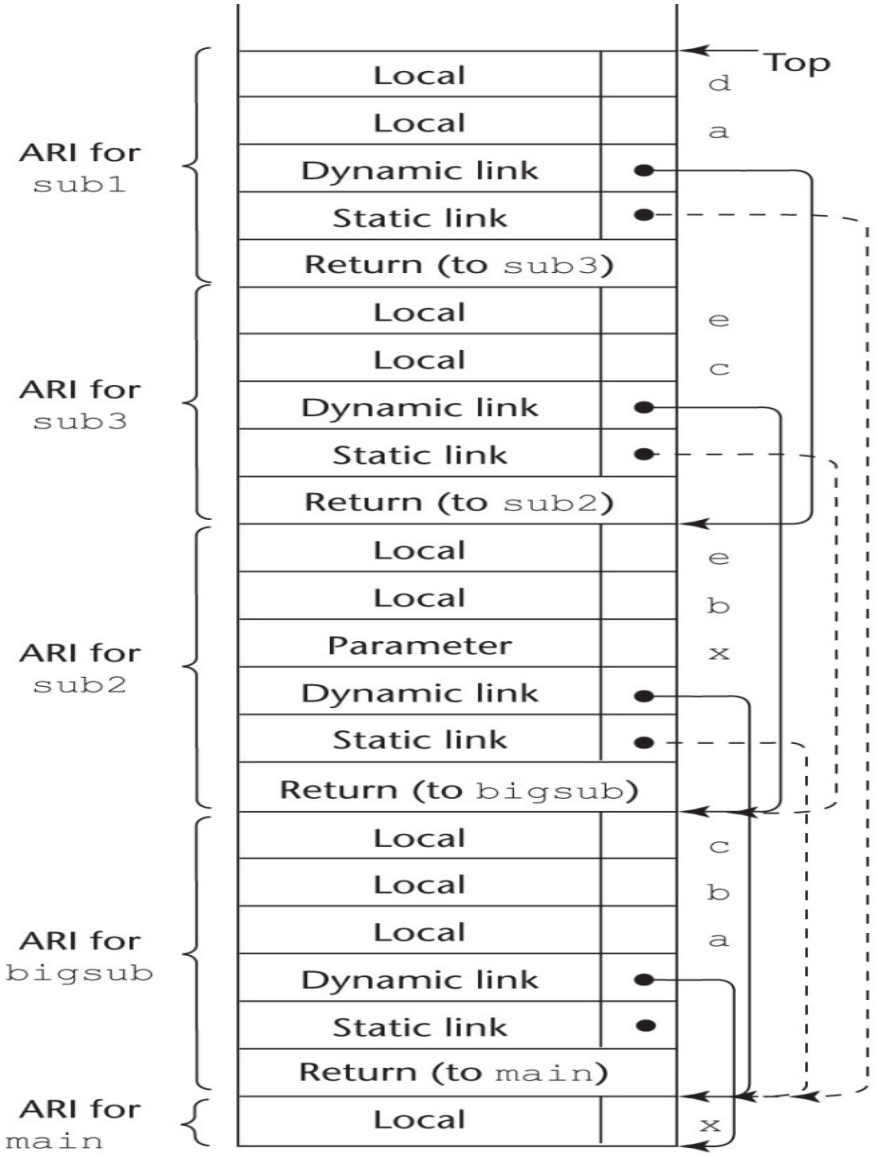
Example JavaScript Program

- Call sequence for `main`

  main calls bigsub
  bigsub calls sub2
  sub2 calls sub3
  sub3 calls sub1

```
function main(){
    var x;
    function bigsub() {
        var a, b, c;
        function sub1 {
            var a, d;
            a = b + c; <--------1
            …
        } // end of sub1
        function sub2(x) {
            var b, e;
            function sub3() {
                var c, e;
                …
                sub1();
                …
                e = b + a; <---2
            } // end of sub3 ...
            sub3();
            …
            a = d + e; <--------3
        } // end of sub2
        …
        sub2(7);
        …
    } // end of bigsub
    …
    bigsub();
    …
} // end of main
```

Stack Contents at Position 1



ARI = activation record instance

main calls bigsub
bigsub calls sub2
sub2 calls sub3
sub3 calls sub1

Static Chain Maintenance

- At the call,
  - The activation record instance must be built
  - The dynamic link is just the old stack top pointer
  - The static link must point to the most recent ARI of the static parent
    - Two methods:
      1. Search the dynamic chain
      2. Treat subprogram calls and definitions like variable references and definitions

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Blocks

- Blocks are user-specified local scopes for variables
- An example in C

```
{int temp;
 temp = list [upper];
 list [upper] = list [lower];
 list [lower] = temp
}
```

- The lifetime of `temp` in the above example begins when control enters the block
- An advantage of using a local variable like `temp` is that it cannot interfere with any other variable with the same name

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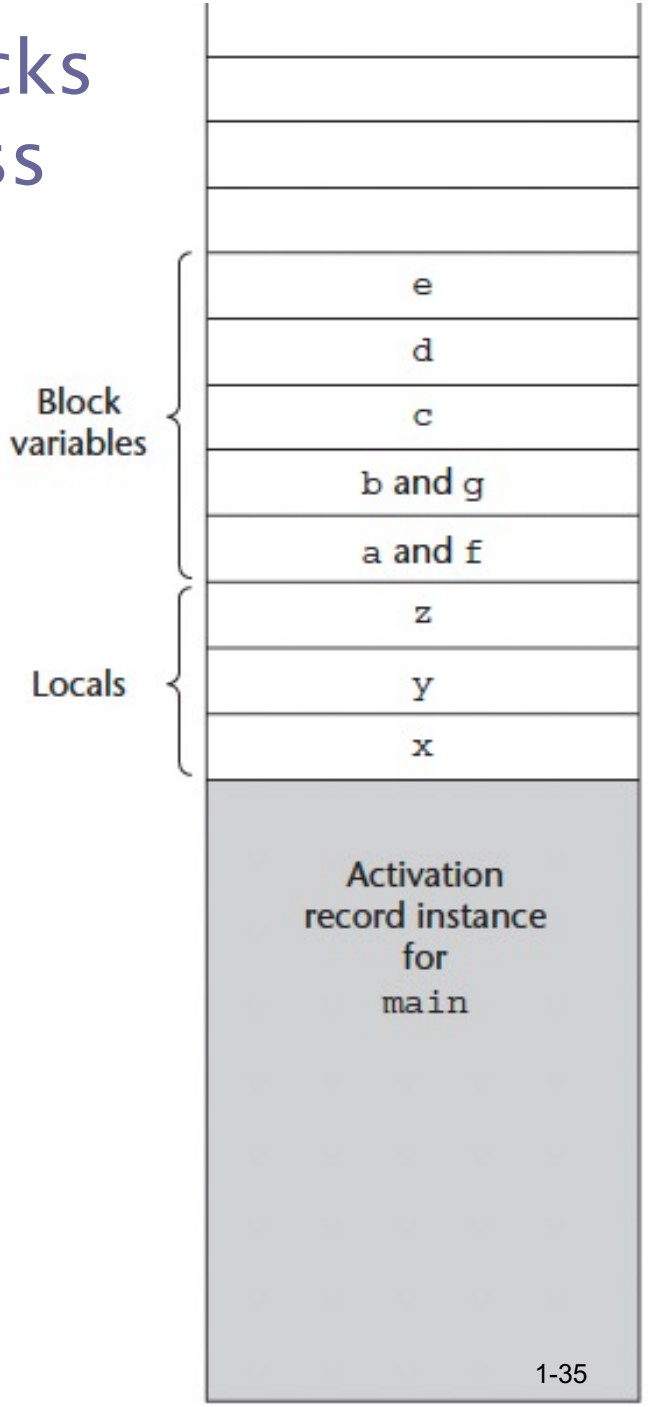
Implementing Blocks

- Two Methods:
  1. Treat blocks as parameter-less subprograms that are always called from the same location
     - Every block has an activation record; an instance is created every time the block is executed
  2. Since the maximum storage required for a block can be statically determined, this amount of space can be allocated after the local variables in the activation record

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Block variable storage when blocks are not treated as parameterless procedures

```
void main() {
  int x, y, z;
  while ( . . . ) {
    int a, b, c;
    . . .
    while ( . . . ) {
      int d, e;
      . . .
    }
  }
  while ( . . . ) {
    int f, g;
    . . .
  }
  . . .
}
```



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Implementing Dynamic Scoping

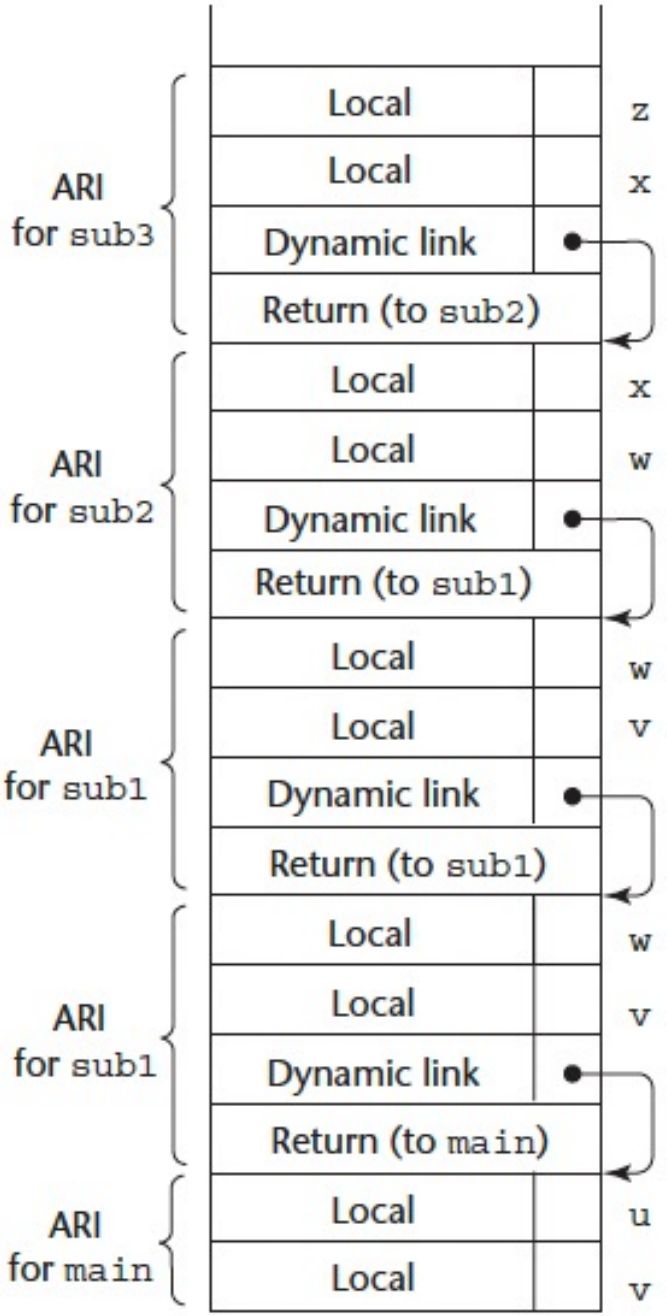
- *Deep Access*: non-local references are found by searching the activation record instances on the dynamic chain
  - Length of the chain cannot be statically determined
  - Every activation record instance must have variable names
- *Shallow Access*: put locals in a central place
  - One stack for each variable name
  - Central table with an entry for each variable name

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Using Deep Access to Implement Dynamic Scoping

```
void sub3() {
  int x, z;
  x = u + v;
  …
}
void sub2() {
  int w, x;
  …
}
void sub1() {
  int v, w;
  …
}
void main() {
  int v, u;
  …
}
```

main	calls	sub1
sub1	calls	sub1
sub1	calls	sub2
sub2	calls	sub3



ARI = activation record instance

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Deep Access vs Static Chain Method

In a dynamic-scoped language,

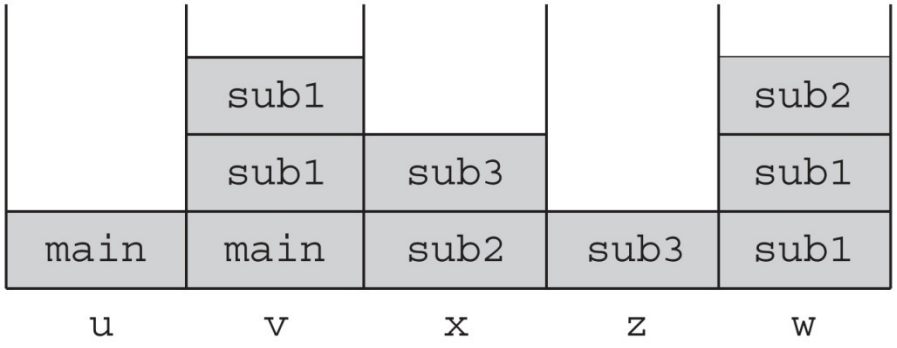
1. There is no way to determine at compile time the length of the chain that must be searched.
   - Every activation record instance in the chain must be searched until the first instance of the variable is found.
   - This is one reason why dynamic-scoped languages typically have slower execution speeds than static-scoped languages.

2. The activation records must store the names of variables for the search process, whereas in static-scoped language implementations only the values are required.

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Using Shallow Access to Implement Dynamic Scoping

```
void sub3() {
  int x, z;
  x = u + v;
  …
}
void sub2() {
  int w, x;
  …
}
void sub1() {
  int v, w;
  …
}
void main() {
  int v, u;
  …
}
```



(The names in the stack cells indicate the program units of the variable declaration.)

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Summary

- Subprogram linkage semantics requires many action by the implementation
- Simple subprograms have relatively basic actions
- Stack-dynamic languages are more complex
- Subprograms with stack-dynamic local variables and nested subprograms have two components
  - actual code
  - activation record

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Summary (continued)

- Activation record instances contain formal parameters and local variables among other things
- Static chains are the primary method of implementing accesses to non-local variables in static-scoped languages with nested subprograms
- Access to non-local variables in dynamic-scoped languages can be implemented by use of the dynamic chain or thru some central variable table method

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